

► BACKGROUND

Globally, more than 55 million people live with dementia [1], and an even greater number of caregivers are also affected by the disease. Wearable technology can offer valuable support in monitoring this health condition. This research continues the development of a wearable device called “HowMI - Home Wearables and Monitors Integrated”, designed to enhance end-user confidence by providing healthcare professionals with an innovative and more accurate care management tool, enabling faster responses in emergency situations [2].

► METHODOLOGY

To support this system-level approach, the overall architecture is illustrated in Figure 1. It highlights the integration between the wearable devices and the supporting technological infrastructure. Data collected by the devices is transmitted wirelessly to base stations or beacons, then relayed to a central artificial intelligence (AI) engine [3]. This engine performs real-time data analysis, enabling anomaly detection and generating alerts. Processed information is securely made available through a web portal accessible to caregivers and health care professionals.

The modular nature of the system ensures that each device can operate independently or in combination, allowing flexible deployment in different care settings, such as private homes or senior residential facilities. The design development of HowMI followed an iterative process guided by user centered and inclusive design principles, with an emphasis on safety, usability, and efficiency for elderly individuals with dementia, hardware and software development proceeded in parallel.

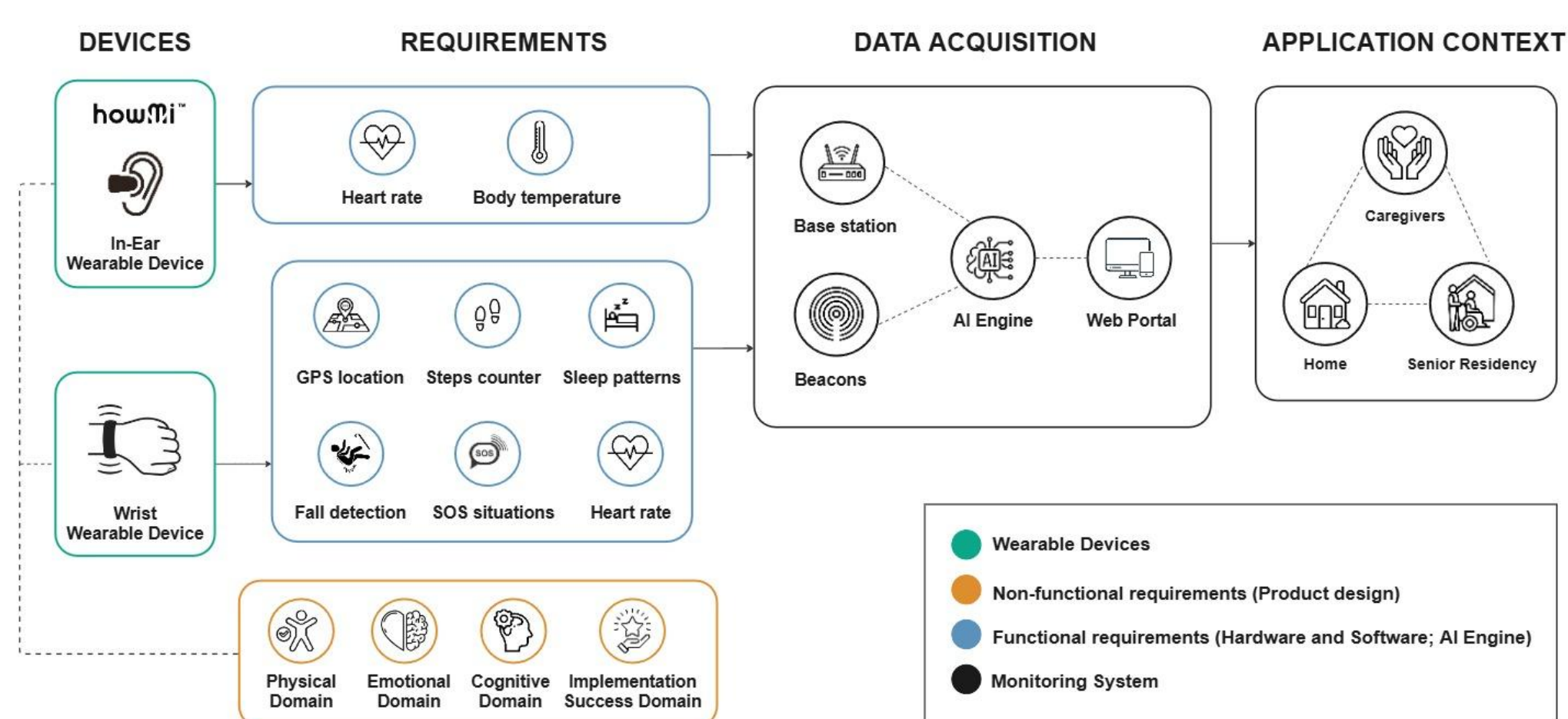


Figure 1. Overview of the project System.

► FUTURE WORK

Future work will focus on advancing the development of the HowMI in-ear wearable device through targeted technical and usability improvements. A key step involves conducting a detailed anatomical study of the ear using 3D models reconstructed from CT scan data. In parallel, a prototype is being developed to begin testing integrated sensors within an in-ear context.

This prototype will allow assessment of sensor performance, focusing on accuracy, signal stability, and sensitivity to movement. Comparative testing of sensing technologies will help identify the most precise, energy-efficient, and mechanically compatible solutions for the ear canal. Further development will address the device’s integration within the broader HowMI system.

► BIBLIOGRAPHY

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► PRELIMINARY STUDIES

PRODUCT DESIGN

To support this research, a systematic literature review was undertaken to explore current trends and challenges in the use of wearable technology for dementia care. The review underscored the emerging potential of ear-based devices such as smart earphones and hearing aids in monitoring key physiological indicators, including heart rate, oxygen saturation, and body temperature. In response to identified limitations in existing solutions, a conceptual design framework was developed, based on user-centered and universal design principles. The resulting requirements were organized into three core domains: physical, emotional, and cognitive. Furthermore, the specific requirements for the HowMI device emphasize the importance of a user-friendly interface, broad adaptability across users, and continuous real-time health data monitoring to enable timely intervention [4].

HARDWARE

The HowMI prototype integrates two key sensors: the MLX90632 infrared sensor, positioned towards the eardrum to measure body temperature, and the MAX30102 PPG sensor, in contact with the ear canal to monitor heart rate and blood oxygen saturation. Both sensors communicate with an ESP32 microcontroller via the I²C protocol, enabling Bluetooth communication for real-time data transmission to external devices.

As a first step in validating the HowMI device, an optimisation of the previously designed prototype was carried out, with particular emphasis on the hardware, aiming to reduce the physical dimensions of the device. At this stage, it was possible to identify certain functional limitations, particularly related to positional and movement artefacts. Small displacements of the device within the ear negatively affect data acquisition quality, potentially leading to inaccurate readings of physiological signals and, consequently, to misinterpretation of the user’s condition.

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